ENGG5301 Information Theory (CUHK) **Fall 2016**

Lecture 11: 21-25 Nov, 2016

Lecturer: Sidharth Jaggi *Scribe: Li Xiaolu*

Disclaimer: *These notes have not been subjected to the usual scrutiny reserved for formal publications. They may be distributed outside this class only with the permission of the Instructor.*

11.1 Definition of security

Security means different things to different people. There is a definition in merriam-webster dictionatry:

`http://www.merriam-webster.com/dictionary/security`

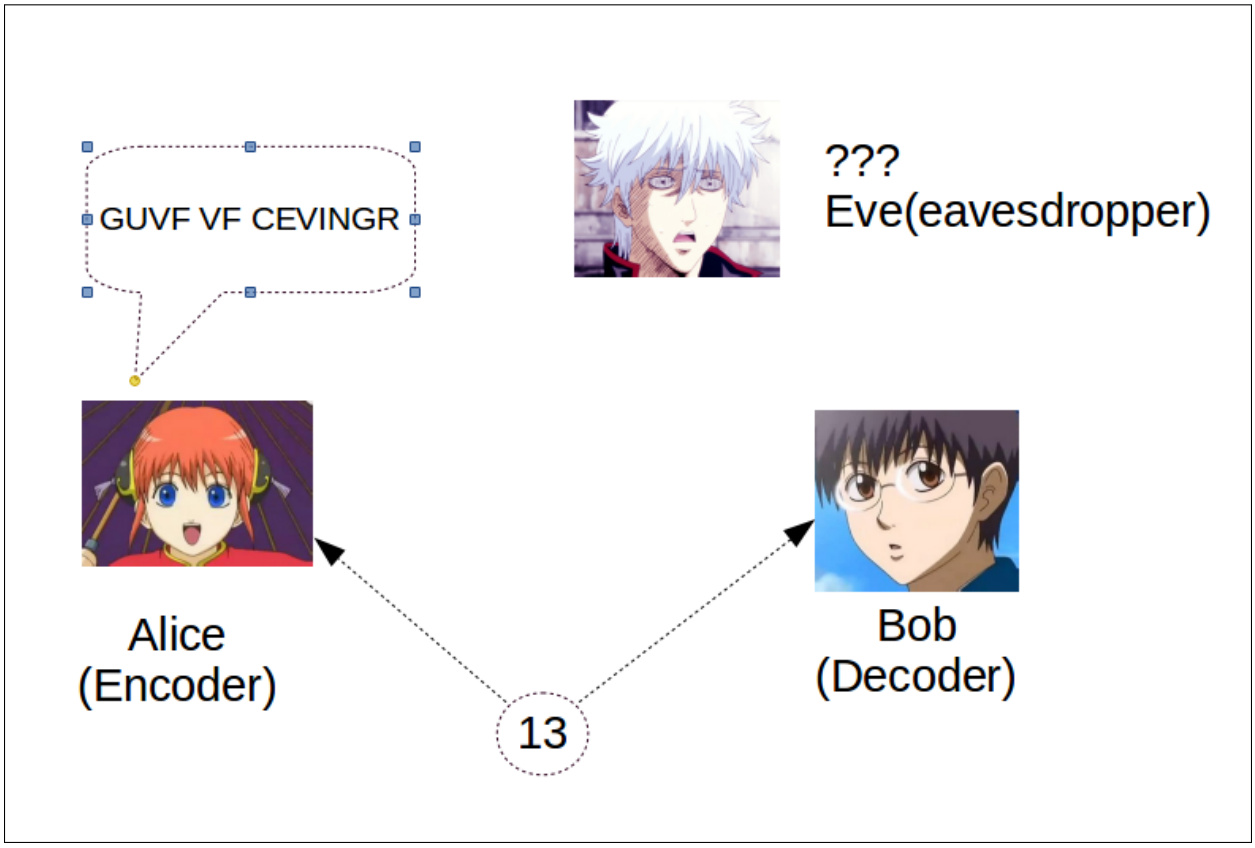
We have several meanings for security:

- Secrecy(privacy).
- Reliability.
- Anonymity.
  - Tor `https://en.wikipedia.org/wiki/Tor_(anonymity_network)`
  - BitCoin `https://en.wikipedia.org/wiki/Bitcoin`
  - ...
- Covertness.
  - Steganography `https://en.wikipedia.org/wiki/Steganography`
  - Digital watermarking `https://en.wikipedia.org/wiki/Digital_watermarking`
  - ...

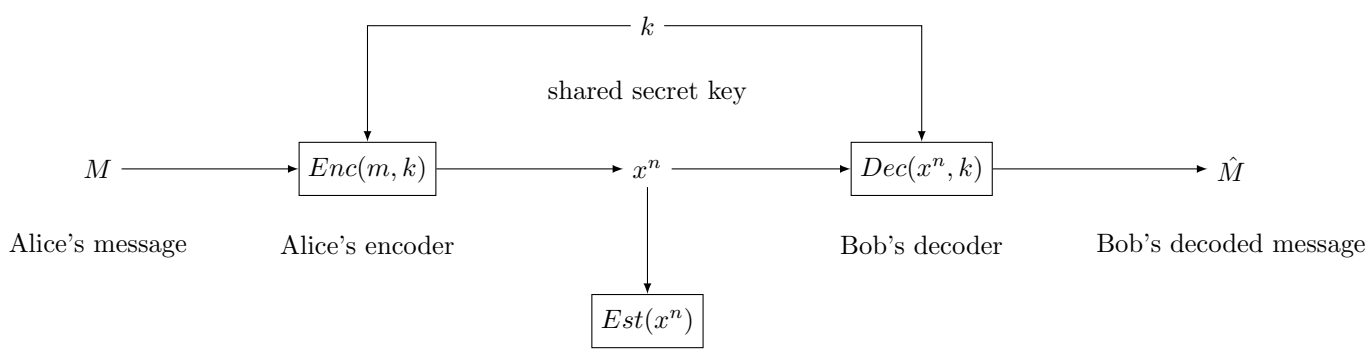
In this course, we will cover Secrecy and Reliability, primary focus on Secrecy. Will not cover Anonymity and Covertness due to time-constraints.

11.2 Eg. ROT13

This is a symmetric-key encryption scheme(shared secret key), and a perfect secrecy scheme example(0-error for decoder)

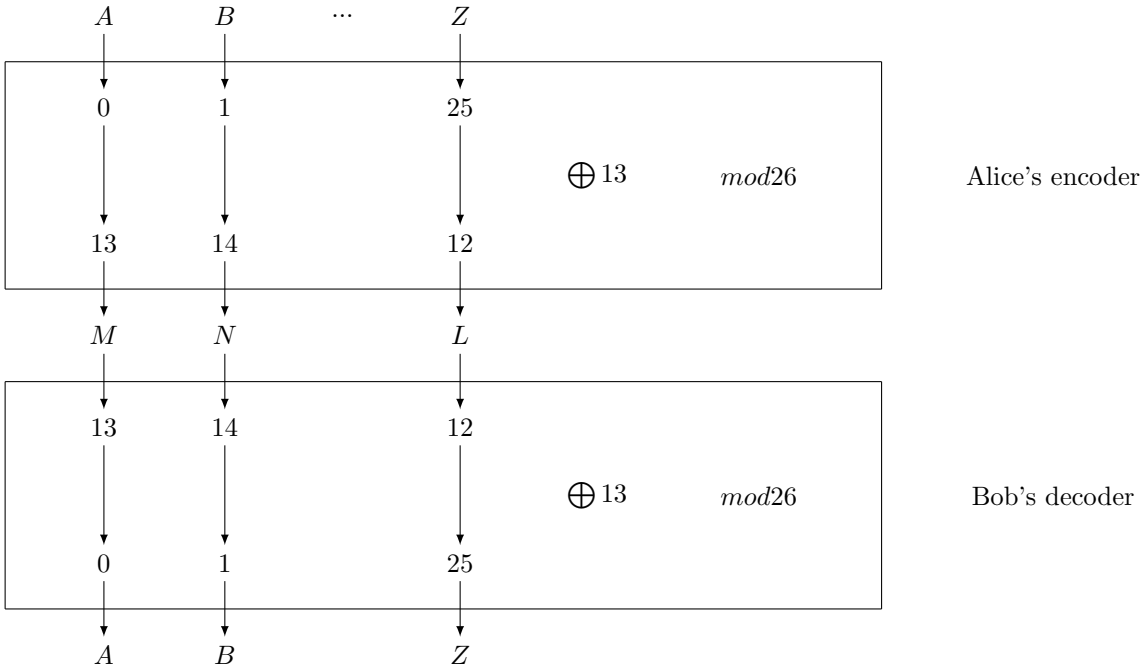


System Diagram



Eg. ROT13: a simple letter substitution cipher that replaces a letter with the letter 13 letters after it in the alphabet.

Goal1: For Bob's decoder, we expect $\forall M, \hat{M} = M$. Because Bob knows the shared key, so we can achieve this goal.



Goal2: For Eve's estimater, we don't want Eve to obtain the original message.

- $M \in [26]^n$
- $x^n \in [26]^n$
- $k \in [26]$
- Eve's estimater knows Enc and Dec, but does not know k

So for Eve, $P_{\hat{M}|x^n}(\hat{m}|x^n) = \frac{1}{26}$ (because $k \in [26]$).

Also, for this example, Eve can use more side information to estimate the message:

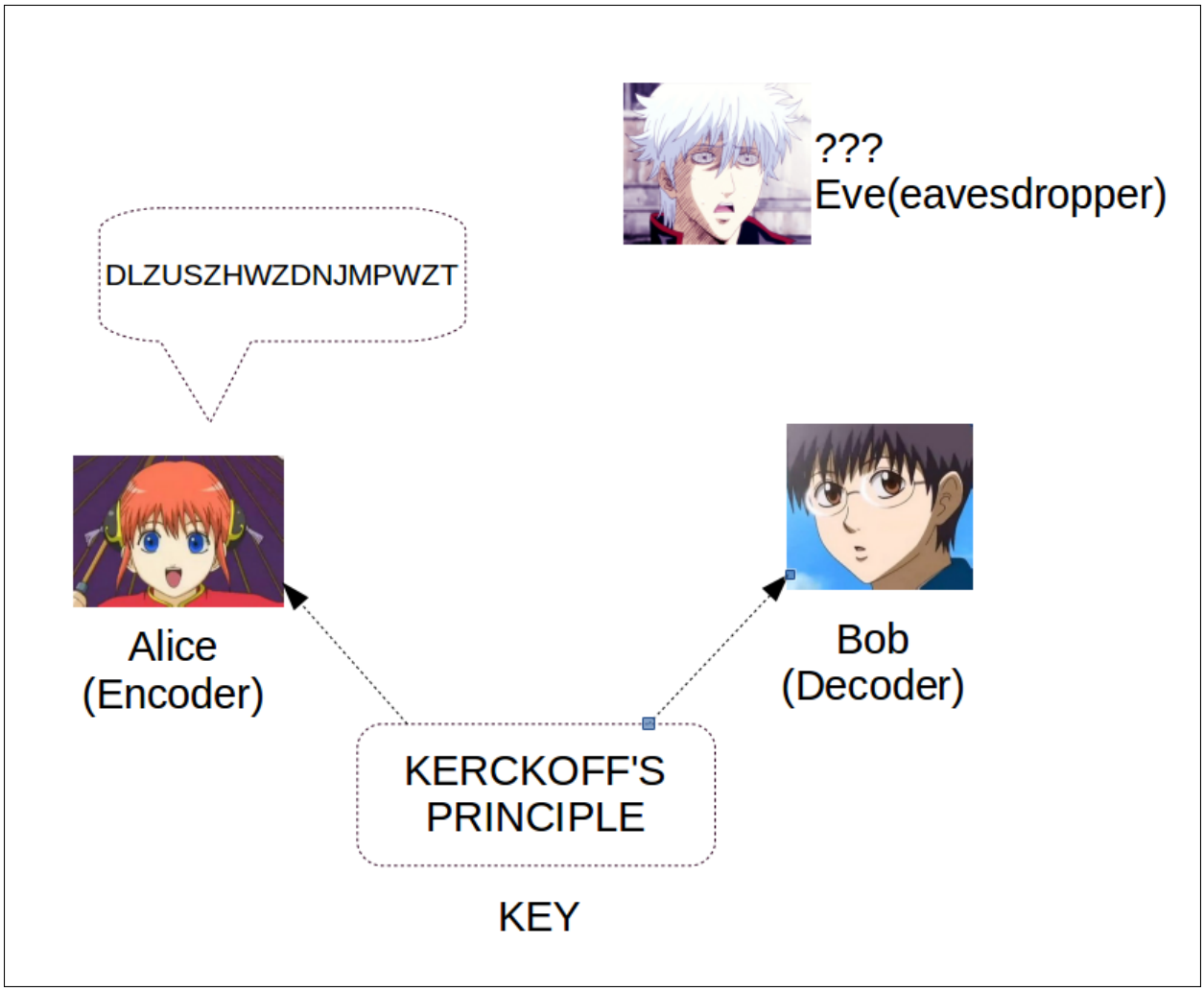
- "VF" appear two times in message
- Eve can also leverage the frequencies of letters in English to help him estimate the original message
- In this example, Eve can choose the most reasonable one message from 26 possible ones he estimated

Using these side information

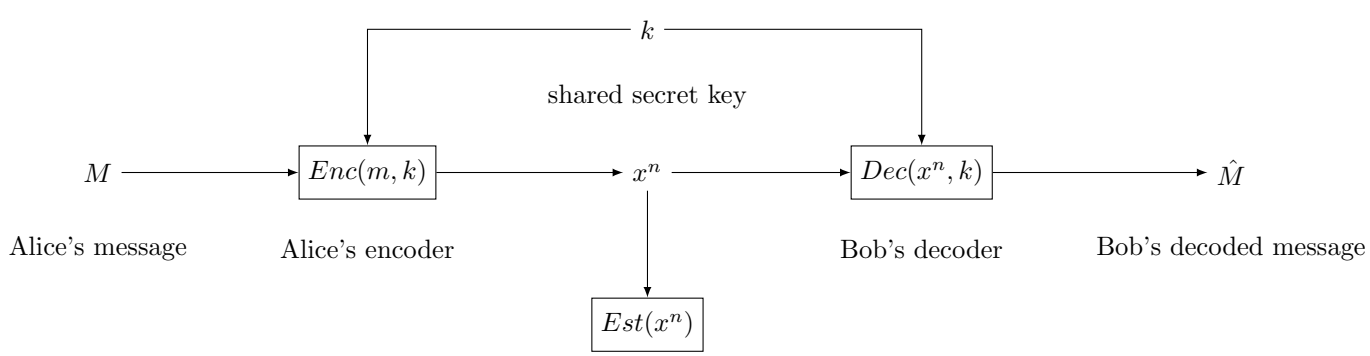
$$P_{\hat{M}|X^n}(\hat{m}|x^n) = 1, \text{ if } \hat{M} = M$$
$$P_{\hat{M}|X^n}(\hat{m}|x^n) = 0, \text{ otherwise}$$

According to Kerckhoff's Principle(1883): A cryptosystem should be secure even if everything about the system, except the key, is public knowledge. So when designing codes, we will need to carefully distinguish between $(Enc(), Dec())$ and k.

11.3 Eg. One-time pads (OTPs)

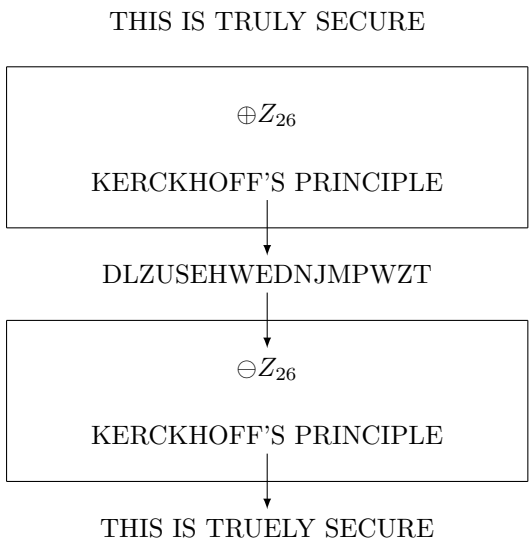


System Diagram



Eg. One Time Pad: an encryption technique that cannot be cracked, but requires the use of a one-time pre-shared key the same size as the message being sent.

Goal1: For Bob's decoder, we expect $\forall M, \hat{M} = M$. Because Bob knows the shared key, so we can achieve this goal.



Goal2: Security against Eve

- $k \in [26]^n$
- $P_{M^n|X^n}(m^n, x^n) = \frac{P_{M^n,X^n}(m^n,x^n)}{P_{x^n}(x^n)} = \frac{P_{X^n|M^n}(x^n|m^n)P_M(m^n)}{\sum_{m'^n} P_{X^n|M^n}(x^n|m'^n)P_M(m'^n)}$ (Bayes Rule)

Note that:

(i) $Enc(m_1^n, k^n) \to m_1^n \oplus_{26} k^n = (m_1 \oplus_{26} k_1, m_2 \oplus_{26} k_2, ..., m_n \oplus_{26} k_n)$

(ii) $P_k(k_i) = \frac{1}{26}$

Therefore:

$$\forall k, x^n, P_{X^n|M^n}(x^n|m^n) = \prod_{i=1}^n P_{X_i|M}(x_i|m) = (\tfrac{1}{26})^n$$

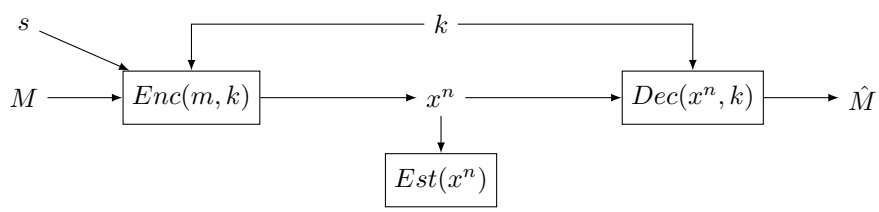
$$\therefore P_{M^n|X^n}(m^n|x^n) = \frac{P_{X^n|M^n}(x^n|m^n)P_M(m^n)}{\sum_{m'^n} P_{X^n|M^n}(x^n|m'^n)P_M(m'^n)}$$

$$= \frac{(\frac{1}{26})^n * P_M(m)}{\sum_{m'} (\frac{1}{26})^n * P_M(m)} = P_M(m)$$

So using this scheme, Eve's estimater cannot improve on prior distribution. But the limit is that all message should have the same length with the shared key.

11.4 Multiple security metrics in the literature

System Diagram



11.4.1 Perfect Secrecy: Symmetric key encryption scheme

(Enc, Dec) is a perfect secrecy scheme if $\forall(M, M'), P_{k,s}(Enc(M, k, s)) = P_{k,s}(Enc(M', k, s))$

- OTPis perfectly secure
- $\forall(m, x^n), P_{M|X^n}(m|x^n) = P_M(m)$
- $I(M : X^n) = 0 \ \because I(M : X^n) = \sum_{m,x^n} p(m, x^n) \log \frac{p(m,x^n)}{p(m)*p(x^n)} = \sum_{m,x^n} p(m, x^n) \log \frac{p_{m|x^n}(m|x^n)}{p_M(m)} = 0$
  Which means x^n reveals no information about the message.

But sometimes demanding perfect secrecy is too much stringent. We also have a "natural" relaxation.

11.4.2 Message indistinguishability

(Enc, Dec) is said to be $\epsilon - message$ indistinguishable if

$$\forall M, M', T : x^n \to 0, 1$$
$$|P_{k,s}(T(Enc(M, k, s)) = 1) - P_{k,s}(T(Enc(M', k, s)) = 1)| < \epsilon$$

Which implies:

- $\forall M, M', |P_{k,s}(Enc(M, k, s)) - P_{k,s}(Enc(M', k, s))| < \epsilon$
  - Choosing T() to be the Indicator function
  - distribution on x^n are very similar, whether m or m' transmitted
- $\forall m, x^n, P_{M|X^n}(m|x^n) \in P_M(m)(1 \pm 2\epsilon)$
  $P_{M|X^n}(m|x^n) = \frac{P_{X^n|M}(x^n|m)*P_M(m)}{\sum_{m'} P_{X^n|M}(x^n|m')*P_M(m')}$
  $\because$ distribution on x^n are very similar given m, we can define $P_{X^n|M}(x^n|m') = (1 \pm \epsilon)P_{X^n|M}(x^n|m)$
  $\therefore \mathtt{P_{M|X^n}(m|x^n)} = \frac{\mathtt{P_{X^n|M}(x^n|m)*P_M(m)}}{\sum_{\mathtt{m'}}(1\pm\epsilon)\mathtt{P_{X^n|M}(x^n|m)*P_M(m')}}$
  $= \frac{P_M(m)}{(1\pm\epsilon)\sum_{m'} P_M(m')} = \frac{P_M(m)}{(1\pm\epsilon)}$

Using taylor series of $\frac{1}{1\pm x} = 1 \pm x + ...$

$\therefore \mathtt{P_{M|X^n}(m|x^n) = (1 \pm \epsilon)P_M(m) \in (1 \pm 2\epsilon)P_M(m)}$

- $I(M; X^n) \leq f(\epsilon)$
  - $I(M; X^n) = \sum_{m,x^n} p(m, x^n) \log \frac{p_{m|x^n}(m|x^n)}{p_M(m)} = \sum_{m,x^n} p(m, x^n) \log 1 \pm 2\epsilon \leq f(\epsilon)$
  - this condition also called "strong secrecy"
  - Early literature in Information Theoretic Secrecy focused on a very weak notion security called "weak secrecy": $I(M; X^n) \leq n\epsilon$

11.5 Computational Security

Computational Security is commonly called "cryptography", which makes assumptions on the types of estimators Eve can use. "reasonable" assumption is that Est(x^n) should be "relatively" computationally efficient. (eg. $poly(n)$)

Encryption scheme (Enc, Dec) is said to be $(t, \epsilon) - message$ indistinguishable, if

$$\forall M, M', T : x^n \to 0, 1$$

$$\text{computational complexity} \leq t$$

$$|P_{k,s}(T(Enc(M, k, s)) = 1) - P_{k,s}(T(Enc(M', k, s)) = 1)| < \epsilon$$

This "small" change has profound implications because two distributions may be very different but may be hard to verify computationally efficiently.

And this is the foundation of much of "practical" cryptography used daily.

However, making the assumption of computational hardness is not always right.

Also, without computational assumption, some things may probably impossible, but with the computational assuption, we can make it possible.

Eg. It is probably information-theoretically impossible to have a public-key scheme that ensures $I(M; X^n) < \epsilon$

However with computational assumptions on Eve's $Est(x^n, p)$, we can get commonly used PR Schemes such as Diffie-Hellman, RSA, ...

11.6 Extra Thinking about OTP scheme

In OTP scheme, the number of bits of M is H(M):

$H(M) = H(M|X^n, K) + I(X^n, K; M)$

$= H(M|X^n, K)$ (①) $+ I(M; X^n)$ (②) $+ I(M; K|X^n)$ (③)

- ①

  Fano's inequality with side-information K:

First We can define a variable E:

$$E = \begin{cases} 1 & \text{if } \hat{M} = M \\ 0 & \text{otherwise} \end{cases} \tag{11.1}$$

Then:
$H(E, M|X^n, K) = H(E|X^n, K) + H(M|E, X^n, K)$:
where $H(E|X^n, K) \leq H(E) = H(P_e) \leq 1$,
$H(M|E, X^n, K) = P(E = 1)H(M|E = 1, X^n, K) + P(E = 0)H(M|E = 0, X^n, K)$
$\because$ When E=1, which means $\hat{M} = M$, $H(M|E = 1, X^n, K) = 0$
$\therefore H(M|E, X^n, K) = P_e H(M|E = 0, X^n, K) \leq P_e H(M)$
$\therefore H(E, M|X^n, K) \leq 1 + P_e H(M)$
Also, we can calculate $H(E, M|X^n, K) = H(M|X^n, K) + H(E|M, X^n, K)$:
where $H(E|M, X^n, K) = 0$ when M, X^n, K are given
$\therefore$ we can get $H(M|X^n, K) \leq 1 + P_e H(M)$

- ②
  According to weak secrecy, we can get $I(M; X^n) \leq n\epsilon$
- ③
  $I(M; K|X^n) = H(K|X^n) - H(M|K, X^n) = H(K) - 0$

$\therefore H(M) \leq 1 + P_e H(M) + n\epsilon + H(K)$

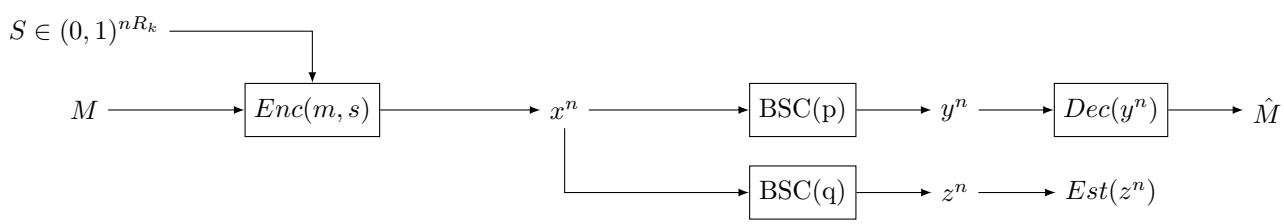
Here, $H(M)$ can be very large, and also by choosing proper Enc & Dec, we can make P_e very small.

So to make the OTP security enough, we should make sure that $H(M) \leq H(K)$.

So each time we should transmit a key as long as the message itself, that's not good. Can we do better?

11.7 The Wiretap Channel

Wiretap Channel



- $S \in (0,1)^{nR_k}$ where R_k is the key rate.
- $M \in (0,1)^{nR}$ where R is message rate.

- Enc is Alice's Encoder
- $x^n \in (0,1)^n$
- BSC(p) is the primary channel
- BSC(q) is the wiretapper channel

This is a wiretap channel, in wiretap channel, there is no common key between Alice and Bob.

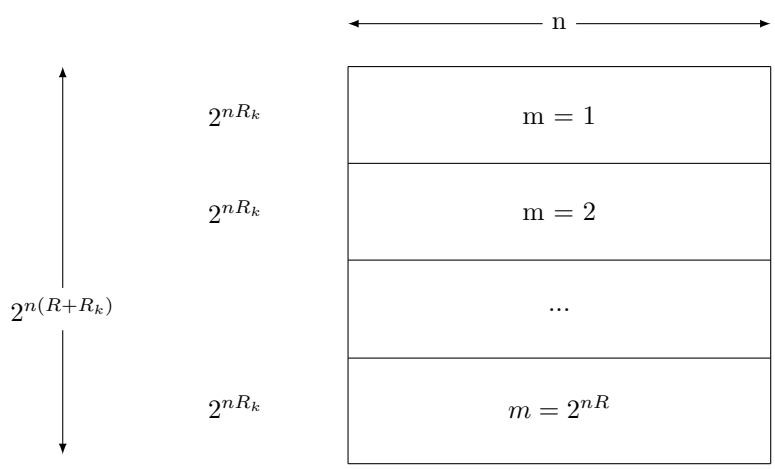
- Goal1 $\forall m, Pr(\hat{M} \neq M) < \epsilon_n \downarrow 0$ for Reliability
- Goal2 $I(M : Z^n) < \epsilon$

11.7.1 Observation

We can only hope for $R > 0$ if $p < q$(when both $< \frac{1}{2}$), which means primary channel is less noisy than wiretapped channel. If not, then Eve can use Bob's decoder as its estimator to decode m, so $H(M; Z^n) = H(M)$

11.7.2 Achievability

We can design the codebook like this $\mathcal{C}$:



- $Enc(m, s)$

  For message m, choose the sth codeword from the mth partition as x^n

- $Dec(y^n)$

  Given y^n, Bob decodes to (hopefully) unique $x^n \in B_H(y^n, np(1+\epsilon_3))$. And then decodes to corresponding M (actually can also decode S, but we doesn't care about that)

  Because Bob decodes to $B_H(y^n, np(1+\epsilon_3))$, so $P_e = \frac{2^{nH(p(1+\epsilon_3))}}{2^n} = 2^{n(H(p(1+\epsilon_3)-1)}$

$\because H(p(1+\epsilon_3)-1<0$

$\therefore P_e$ can be small enough. So Bob can correctly decode to x^n.

And thus can correctly decode to M because the code book is shared by Enc and Dec.

I'm not sure whether the demonstration is right.

11.7.3 Security

Our goal to prove security is that we want to show $I(M:Z^n)<\epsilon$ which is small enough.

- $I(X^n;Z^n) \geq I(M;Z^n)$
  $\because I(X^n;Z^n) = H(Z^n) - H(Z^n|X^n)$ $I(M;Z^n) = H(Z^n) - H(Z^n|M)$
  and $H(Z^n|M) > H(Z^n|X^n)$, so $I(X^n;Z^n) \geq I(M;Z^n)$.
  problem $\rightarrow$ If we can show $I(X^n;Z^n)<\epsilon$, then problem solved.
- $I(X^n;Z^n|D) \geq I(X^n;Z^n)$ if we reveal to Eve $D = d_H(z^n,x^n)$ (bonus question).
  so problem $\rightarrow$ to show $I(X^n;Z^n|D)<\epsilon$
- to show $I(X^n;Z^n|D)<\epsilon$. We first define an event T:

$$T = \begin{cases} 1 & \text{if } D \in nq(1\pm\epsilon) \\ 0 & \text{otherwise} \end{cases} \tag{11.2}$$

  $\therefore I(X^n;Z^n|D) = \sum_{x^n,z^n} p(x^n,z^n)\log\frac{p(x^n,z^n)}{p(x^n)p(z^n)}$
  $= \sum_{x^n,z^n D\in nq(1\pm\epsilon)} p(x^n,z^n)\log\frac{p(x^n,z^n)}{p(x^n)p(z^n)} + \sum_{x^n,z^n D\notin nq(1\pm\epsilon)} p(x^n,z^n)\log\frac{p(x^n,z^n)}{p(x^n)p(z^n)}$
  $= I(X^n;Z^n|D,T=1) + I(X^n;Z^n|D,T=0)$
  According to chernoff bound: $Pr(X \notin \mu(1\pm\delta)) \leq e^{-\frac{\delta^2\mu}{3}}$
  $\therefore Pr(T=0) \leq e^{-nq\epsilon^2} \leq 2^{-nq\epsilon^2}$
  $Pr(T=1) \geq 1-2^{-nq\epsilon^2}$
  $\therefore I(X^n;Z^n|D,T=0) \leq Pr(T=0)nI(X;Z) \leq 2^{-nq\epsilon^2}nI(X;Z) \leq 2^{-nq\epsilon^2}nH(X) \leq 2^{-nq\epsilon^2}n\log\mathcal{X} \leq n2^{-nq\epsilon^2}$ which is negligible
  so problem $\rightarrow$ to show $I(X^n;Z^n|D,T=1)<\epsilon$
- to show $I(X^n;Z^n|D,T=1)<\epsilon$
  equals to show $= \sum_{x^n,z^n D\in nq(1\pm\epsilon)} p(x^n,z^n)\log\frac{p(x^n,z^n)}{p(x^n)p(z^n)} < \epsilon$
  $\sum_{x^n,z^n D\in nq(1\pm\epsilon)} p(x^n,z^n)\log\frac{p(x^n,z^n)}{p(x^n)p(z^n)} = \sum_{x^n,z^n D\in nq(1\pm\epsilon)} p(x^n,z^n)\log\frac{p(x^n|z^n)}{p(x^n)}$
  $\because \frac{p(x^n|z^n)}{p(x^n)} = (1\pm\delta)$
  $\therefore I(X^n;Z^n|D,T=1) = \sum_{x^n,z^n D\in nq(1\pm\epsilon)} p(x^n,z^n)\log(1\pm\delta) \leq \log(1\pm\delta) = \epsilon$
  so security is proved.
- In notes, there is a possible pair of (R,R_k):

$$R = H(q) - H(p) - \epsilon_1$$

$$R_k = 1 - H(q) + \epsilon_2.$$

We can similarly calculate $I(X^n; Z^n|D, T = 0)$ which can suffice the reliability as well as the security

For any fixed message m', expected $\#codewords \in Sn(z^n, D) = \frac{Vol(Sn(z^n,D))}{2^n} * x^{nR_k}$

$= \frac{2^{nH(D/n)}}{2^n} * x^{n(1-H(q)+\epsilon_2)}$

$\because D \in nq(1 \pm \epsilon)$

$\therefore H(D/n) = H(q(1 \pm \epsilon))$

for $H(q(1+x))) = -q(1+x)\log q(1+x) - (1-q(1+x))\log(1-q(1+x)) \triangleq f(x)$

then $f(0) = H(q)$

$f'(x) = -q\log q(1+x) * q\frac{-q(1+x)}{q(1+x)} + q\log(1-q(1+x)) * (-q)\frac{-(1-q(1+x))}{(1-q(1+x))}$

$= q^2\log q(1+x) + q^2\log(1-q(1+x))$

$\therefore f'(0) = q^2\log q(1-q)$

$\therefore f(x) \geq H(q) - f'(0)x$

$\therefore H(q(1 \pm \epsilon)) \geq H(q) - \frac{\epsilon_2}{2}$

$\therefore \#codewords \geq 2^{n\frac{\epsilon_2}{2}}$

We can choose any $value < 2^{n\frac{\epsilon_2}{2}}$ say $2^{n\frac{\epsilon_2}{3}}$ using chernoff bound:

$\delta = \frac{2^{n\frac{\epsilon_2}{3}}}{2^{n\frac{\epsilon_2}{2}}} = 2^{-n\frac{\epsilon_2}{6}}$

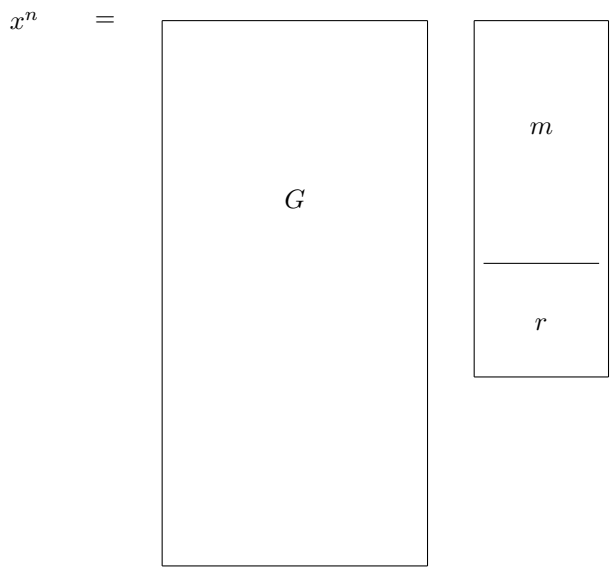
$Pr(X \notin 2^{n\frac{\epsilon_2}{2}}(1 \pm \delta)) \leq e^{-\frac{1}{3}\delta^2\mu}$

so $I(X^n; Z^n|D, T = 0)$ is negligible for the same reason.

Then we can prove security the same as before.

11.8 Example: Linear Codes

11.8.1 Construct Codebook



- $m \in (0,1)^{nR}$
- $r \in (0,1)^{nR_k}$
- $G_{n*n(R+R_k)}$

11.8.2 Reliability

Let's look at any $\bar{x}^n$ and analyze the $Pr(\exists(\bar{m},\bar{x})G(\bar{m}\bar{x})^T = \bar{x}^n)$

$\because G$ is random codebook,

$\therefore Pr(G(\bar{m}\bar{x})^T)) = \frac{1}{2^n}$

$\therefore Pr(\exists(\bar{m},\bar{x})G(\bar{m}\bar{x})^T = \bar{x}^n) \leq 2^{n(R+R_k-1)}$

$\because \#\bar{x}^n \in B_H(y^n, np(1+\epsilon)) \doteq 2^{nH(p(1+\epsilon))}$

$\therefore P_e \leq 2^{n[R+R_k-1+H(p(1+\epsilon))]}$

By choosing proper R and R_k, we can achieve the goal that P_e is small enough.

Say $R + R_k < 1 - H(p)$

or $R + R_k = 1 - H(p) + \epsilon_2 - \epsilon_1$

$R = H(q) - H(p) - \epsilon_1$

$R = 1 - H(q) + \epsilon_2$

11.8.3 Secrecy

For a random linear code, we can denote vectors on $Sn(z^n, D)$ by $v_i, i \in [1, \binom{n}{D}]$

We can define event $E_{i,m} = 1$ (r st. $G(\bar{m}\bar{r})^T) = v_i$), means now we observe vector v_i on S_n, if the codeword is of message m, then set E to 1.

As the codeword of all messages on S_n is uniform distributed, so $Pr(E_{i,m} = 1) = \frac{2^{nR_k}}{2^n}$, which is the same as we choose a codeword in the codebook, the probability of the codeword from message m.

$\therefore \mathbb{E}(E_{i,m}) = 1 * \frac{2^{nR_k}}{2^n} + 0 * .. = 2^{nR_k - n}$

$\mathbb{E}(E_{i,m}^2) = 2^{nR_k - n}$

$Var(E_{i,m}) = 2^{n(R_k-1)}(1 - 2^{n(R_k-1)})$

We can define $A(D, m, z^n) = \#$ codewords for m at distance D from z^n

$\therefore \mathbb{E}(A(D, m, z^n)) = \binom{n}{D} 2^{n(R_k-1)} = 2^{nH(\frac{D}{n})} * 2^{n(R_k-1)} = 2^{n(H(q(1\pm\epsilon)+R_k-1)}$

$Var(A(D, m, z^n)) = \binom{n}{D} Var(E_{i,m}) = 2^{nH(\frac{D}{n})} * 2^{n(R_k-1)}(1 - 2^{n(R_k-1)}) = 2^{n(H(q(1\pm\epsilon)+R_k-1)}(1 - 2^{n(R_k-1)})$

we can show $\sum_{x^n, z^n D \in nq(1\pm\epsilon)} p(x^n, z^n) \log \frac{p(x^n, z^n)}{p(x^n)p(z^n)} + \sum_{x^n, z^n D \notin nq(1\pm\epsilon)} p(x^n, z^n) \log \frac{p(x^n, z^n)}{p(x^n)p(z^n)}$

$= I(X^n; Z^n | D, A = \mu) + I(X^n; Z^n | D, A \neq \mu)$ instead.

The same logic to prove security as that in wiretap channel, but this time we use chebyshev's inequality.

$Pr(|\#x^n - \mu| > \epsilon) \leq \frac{Var}{\epsilon^2} = \frac{2^{n(H(q(1\pm\epsilon)+R_k-1)}(1-2^{n(R_k-1)})}{\epsilon^2}$

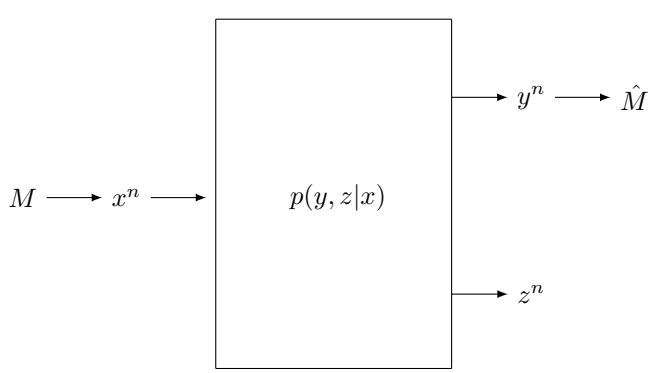
$< \frac{2^{n(H(q(1\pm\epsilon)+R_k-1)}}{\epsilon^2} < \frac{2^{n(H(q)-\epsilon_1+R_k-1)}}{\epsilon^2}$

Choosing $R_k = 1 - H(q) + \epsilon_1$, we can bound to a very small value.

Also for the same demonstration logic above, we can prove secrecy.

11.8.4 Remarks

11.8.4.1 general discrete memoryless channel



By adding a auxiliary R.V. Secrecy Capacity $R = \max_p(u, x)[I(U; Y) - I(U; Z)]$

- Codebook
  - (a) Generate $2^{n(R+R_k)}$ u^n according to $p^*(u)$ which is the optimizing distribution, $R_k = I(U;Z)$
  - (b) Partition into 2^{nR} "bins" of size 2^{nR_k}, each $B(i)$ corresponds to message m_i
- Encoder
  - (a) For message m_i, $u^n \quad U(B(i))$
  - (b) Generates $x^n(u^n)$ randomly according to $p *_{x|u} (x^n|u^n)$ (extra noise to confuse Eve) and transmits this x^n
- Decoder

  Looks for (hopefully) unique u^n s.t. $(u^n, y^n) \in A_\epsilon^{(n)}(u^n, y^n)$
- Eve's estimator

  Since effective channel $u^n \to z^n$ has mutual information $nI(u;z)$, and Bob has $\forall m_i 2^{n(I(u;z)+\epsilon)}$

  So $\forall$ m_i, there are roughly equal $\#u^n$ in each likely type-class for Eve

11.8.4.2 converse

Via a bunch of information-theoretic inequalities. Please refer to EK for details.

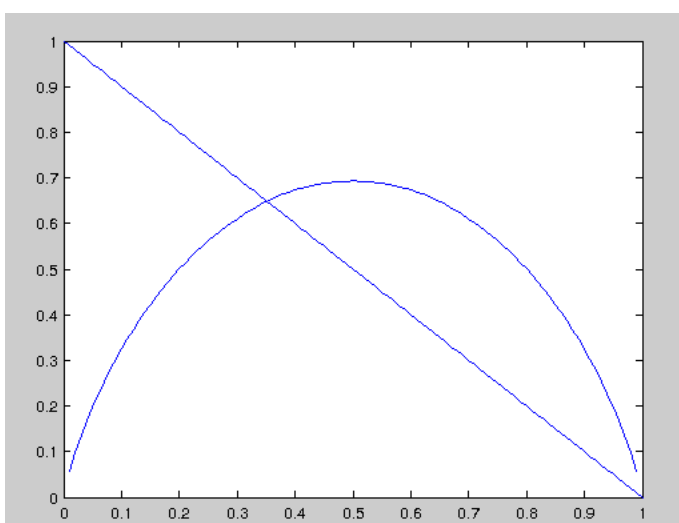
11.8.4.3 special cases

(a) For some $p(y,z|x)$, $\max_p(u,x)[I(U;Y) - I(U;Z)]$ is strictly larger than $\max_{p(x)}[I(X;Y) - I(X;Z)]$

Eg. $X \to Y$ is BSC(p) $I = 1 - H(p)$

$X \to Z$ is BEC(q) $I = 1 - q$

Using matlab to plot the two figures:



If p and q are small enough, $1 - H(p) - (1 - q) < 0$

(b) If Z is a "stochastically degraded" version of y, i.e $p(y, z|x) = p(y|x)p(z|y)$

then $I(U;Y) - I(U;Z) = I(U;Y|Z) = I(U;Y|Z) \leq I(X;Y|Z) = I(X;Y) - I(X;Z)$

so $\max_p(u, x)[I(U;Y) - I(U;Z)] = \max_{p(x)}[I(X;Y) - I(X;Z)]$

Eg. $X \to Y$ is BEC(p), $X \to Z$ is BSC(q)

$$p(y|x) = \begin{array}{ccc} & x{=}0 & x{=}1 \\ y{=}0 & 1-p & 0 \\ y{=}1 & 0 & 1-p \\ y{=}E & p & p \end{array}$$

$$p(z|x) = \begin{array}{ccc} & x{=}0 & x{=}1 \\ y{=}0 & 1-q & q \\ y{=}1 & q & 1-q \end{array}$$

Then we can calculate a virtual matrix, there is one possible solution is:

$$p(z|y) = \begin{array}{ccccc} & z{=}0 & z{=}1 & z{=}E \\ y{=}0 & \frac{1-2p}{1-p} & 0 & \frac{q}{p} \\ y{=}1 & 0 & \frac{1-2p}{1-p} & \frac{q}{p} \end{array}$$

11.8.4.4 AWGN wiretap channel

$Y = X + E_1$ where $E_1 \quad \mathcal{N}(0, N_1)$

$Y = X + E_1$ where $E_1 \quad \mathcal{N}(0, N_2)$

capacity $= \frac{1}{2}\log(1 + \frac{P}{N_1}) - \frac{1}{2}\log(1 + \frac{P}{N_2})$

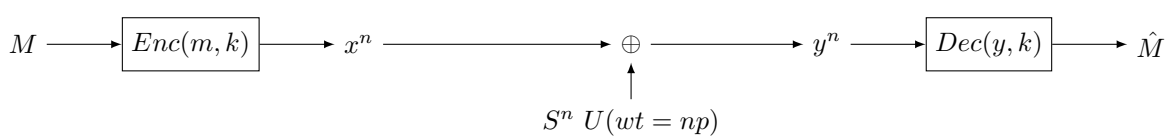
11.8.4.5 Public discussion helps

In addition to wiretapped channel, there is a "public" channel that is noiseless to al parties. (I don't understand this example well)

11.9 Reliability against adversarial jamming

Let's first look at this model, which will have the same result with BSC(p), capacity $= 1 - H(p)$

This is a Bit-flip adversary with common randomness



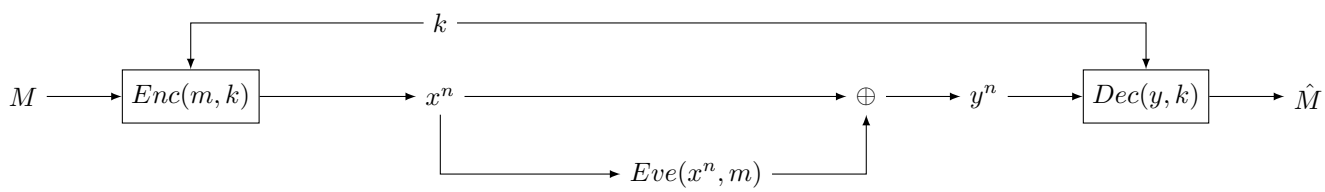
If we add an error vector to x^n, with the weight of np, then this channel act like a BSC(p).

Say if we can add extra randomness to the codeword and can still achieve the reliability and can confuse Eve, that's good.

Then the number error vector can choose from the U(permutation(wh(np))), we can analysis how many bits will it take to add such enough randomness.

$\because \log n! = \sum_{m=1}^{n} \log m \leq \int \log x dx = x \log x - x + c$, then choose shared key with length of $c \log n$ (not sure whether this is right)

System Diagram



We can create codebook like this:

- $R = 1 - H(p) - \frac{c \log n}{2^n}$
- $k \in (0,1)^{c \log n}$ and we can take $c = 10$ as an example, say $k = (s_1, s_2)$, $s_1 \in (0,1)^{5 \log n}$, $s_2 \in (0,1)^{5 \log n}$
- codebook has $2^{nR + \frac{c}{2} \log n}$ codewords

11.9.1 Encoder

Let $q = 2^{\frac{c}{2} \log n}$ $S_1 \in \mathbb{F}_q$ $m \in \mathbb{F}_q^{\frac{nR}{\log q}}$

Define hash $h = \sum_{i=0}^{\frac{nR}{\log q}} m_i s_1^i + s_2$

Now we transmit $x^n(m, h)$

11.9.2 Decoder

(a) First, Bob list-decodes x^n down to list $\mathcal{L}$ of size $l = O(\frac{1}{\epsilon})$, We have shown this in HW4 non-textbook question.

(b) Then, Bob knows s_1 and s_2, so he can know whether there is unique output in the list.

11.9.3 Analysis

(a) Decoding error: if there exists another h and m st. $h - \sum mi(s_1)^i = h' - \sum m_i'(s_1)^i$

Then there will be error. (from the equation we can know if s_1 is the root, then this can be true)

(b) From Eve's perspective, $P_{s_1|x^n}$ $U(\mathbb{F}_q)$. But even he knows h and m, there exists many pairs of (s_1, s_2) which can satisfy the requirements.

Then we want to show P_e is small enough regardless choice of list

$Pr_s 1((h', m') \in \mathbb{L} \quad st \quad error \quad equation \quad satisfied) \leq \frac{nR/\log q}{q}$

(According to Schartz-Zippel lemma, we can calculate this probability)

$\therefore P_e(\exists (h', m') \in \mathbb{L} st \quad error \quad equation \quad satisfied) \leq \frac{|\mathbb{L} nR|}{q \log q}$ where $q = n^{\frac{c}{2}}$

So P_e is small enough.

References

[1] *http://sensor.ee.tsinghua.edu.cn/NIT_Lecture_Notes.pdf*

[2] *https://en.wikipedia.org/wiki/Main_Page*

Thanks for the help of Sid and Eric!

(Not sure whether there are still any problems in it, if it does, please tell me)